MIL-C-46273A(MU)
7 April 1972
SUPERSEDING
MIL-C-46273(MU)
29 August 1960

MILITARY SPECIFICATION

CARTRIDGE, DELAY: M70, ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Delay: M70, Assembly, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MS-28741	- Hose Assembly, Detachable End-Fitting, Medium Pressure

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command

D8593274	- Cartridge, Delay; M70, Assembly
B8799621	- Carton, Packing, Ammunition For Cartridge, Delay, M70
B8799620	- Box, Packing, Ammunition For Cartridge Delay, M70
IEL-8593274	- Inspection Equipment List for Cartridge, Delay: M70, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, (ANSI), 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.-The cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the delay charge, primer, igniter or propellant or derangement of components.

3.4 Primer sensitivity.-The primer shall fire when drop tested at an energy level of 26 inch ounces in a sectioned cartridge case, void of delay charge, igniter and propellant charge.

3.5 Functional.

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3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.-The cartridge when tested in an M10 and M30A1 Initiator shall produce pressures and delay times at the temperature specified in Table I.

TABLE I

	Pressure (psi) 1/		Delay Time, Seconds	
Temperature	Minimum	Maximum	Minimum (see 6.2)	Maximum
-65° to -70°F	500	1500	--	2.50
65° to 75°F	525	1675	1.80	2.20
160° to 165°F	600	1675	1.60	--

1/ Pressure values shall be produced in a one cubic inch pressure station at the end of 15 feet of hose (Size No. 4) in accordance with MS 28741.

3.6 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

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4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 50 cartridges for primer sensitivity, 5 cartridges for sealant tests (to be selected from primer sensitivity sample), 30 cartridges for ballistic tests, 12 for vibration tests (to be selected from ballistics sample) and 10 sets of unassembled components for dimensional and visual examination and tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawing and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

TABLE II

Temperature	Vibrated	Not Vibrated
-65° to -70°F	4	6
65° to 75°F	4	6
160° to 165°F	4	6

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

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4.3.1.2 Size of lot.-An ammunition lot of cartridges shall consist of an identifiable group of cartridges assemblies consisting of:

a. Components produced by a homogeneous process.
b. Delay elements from not more than one lot.
c. Primers from not more than one lot.
d. Igniters from not more than one lot.
e. Propellant from not more than one lot.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables III and IV. For an individual defect (single characteristic) the criteria in Table III shall be used. For total defects within a major or minor class, the criteria in Table IV shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ANSI B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber "O" rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE III

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22,000	300	4	5	5	6

TABLE IV

Sample Size for Major Defects Number of Defects in Class	Total Allowable Defects Per Sample Size			
	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
20 - 43	4	5	8	10
Over 43	6	8	11	14

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4.3.2 (Cont'd)

Sample Size for Major Defects Number of Defects in Class	Total Allowable Defects Per Sample Size 110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Sealing disc, Drawing C8594077, covering a detail of Drawing D8593274.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Diameter of body, min.	Gage
102. Thickness	Gage
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Varnish missing or inadequate	Visual

4.3.2.2 Case, cartridge, Drawing C8595202, part No. 8595202 covering a detail of Dwg. D8593274.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage

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4.3.2.2 (Cont'd)

Defect	Method of Inspection
MAJOR	
103. Diameter of body, min.	Gage
104. Diameter of head, min.	Gage
105. Thickness of base	Gage
106. Thickness of wall	Gage
107. Location of datum diameter	Gage
108. Protective finish inadequate or defective	Visual
109. Burr or sharp corner	Visual
110. Foreign matter	Visual
111. Hardness	Gage
MINOR	
201. Diameter of head, max.	Gage
202. Diameter of body, max.	Gage
203. Total length	Gage
204. Radius or chamfer missing or incorrect	Visual
205. Surface finish improper	Visual

4.3.2.3 Delay element M5, Drawing 8594080 covering a detail of Drawing D8593274.

CRITICAL	
1. Mixed delay elements	Visual
2. Flash hole missing or obstructed	Visual
3. Primer uncrimped or seal missing	Visual
4. Marking missing, incorrect or illegible	Visual
5. Delay charge not visible through flash hole	Visual
6. Metal defective	Visual
MAJOR	
101. Protective finish inadequate or defective	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined	

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4.3.2.4 Cartridge, Delay, M70, Drawing D8593274.

Defect	Method of Inspection
CRITICAL	
1. Any part missing	X-ray & Visual 1/
2. Improper assembly	X-ray & Visual 1/
3. Mixed delay element or propellant	Visual
4. Metal defective	Visual
5. Crimp missing or incorrect	Visual
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Cartridge profile failure	Gage
102. Diameter of body, max.	Gage
103. Diameter of head, max.	Gage
104. Location of datum diameter	Gage
105. "O" Ring split, cracked or cut	Visual
106. Protective finish inadequate or defective	Visual
107. Foreign matter	Visual
108. Burr	Visual
MINOR	
201. Total length	Gage
202. Case damaged	Visual
203. Varnish missing or inadequate	Visual

1/ The cartridge shall be x-rayed with the long axis of the case perpendicular to the rays. The x-ray negatives properly identified, shall be forwarded to the procuring activity.

4.3.2.5 Carton, Packing, Ammunition for Cartridge Delay M70, Drawing B8799621.

CRITICAL

None defined

MAJOR

None defined

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4.3.2.5 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Separators or fillers missing	Visual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual

4.3.2.6 Box, Packing, Ammunition for Cartridge, Delay M70, Drawing B8799620.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual & Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts and articles submitted by a supplier shall be in accordance with provisions set forth in MIL-STD-105 and MIL-C-60539.

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4.3.3.2 Vibration.-Twelve of the 30 assembled x-rayed cartridges of the initial production sample shall be subjected to vibration in accordance with 3.3. After vibration the 12 vibrated cartridge shall be x-rayed as directed in note 1/ 4.3.2.4. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot. When there is a change in the method of charging the cartridge or in the lot of primers, 12 additional cartridges shall be selected and tested as indicated herein.

4.3.3.3 Airtightness.-Each cartridge shall be subjected to an airtightness test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.4 Primer sensitivity.-A sample of 50 assembled cartridges from the initial production sample and 25 assembled cartridges from each subsequent production lot shall be tested for sensitivity. Failure of one cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.5 Sealant application.-A sample of 5 cartridges (consumed in the primer sensitivity test) shall be disassembled and examined for improper sealant application. Failure of any cartridge to comply with the drawing requirements shall be cause for rejction of the lot.

4.3.3.6 Hardness.-Fifty units from each heat treated lot shall be selected for hardness tests. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

4.3.3.7 Functional.-The following tests shall be performed by the government.

4.3.3.7.1 Ballistics.-Thirty test cartridges from each production lot shall be grouped and temperature conditioned in accordance with 4.4.7.1.1 and Table V. Failure of any cartridge to comply with the requirements of 3.5.2 shall be cause for rejection of the lot.

TABLE V

Temperature	Sample Size
-65° to -70°F	10
65° to 75°F	10
160° to 165°F	10

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL 8593274.

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4.4 Test methods and procedures.

4.4.1 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge cases shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.2 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Airtightness.-The method and procedure for performing the airtightness test shall be by use of equipment listed on IEL 8593274.

4.4.4 Primer sensitivity.-Remove and discard the charged delay body and all powder charges from the test cartridges by sectioning the cartridge case around its periphery, one inch, maximum, above the base. Drop test 50 sectioned cartridges from the intial production sample and 25 sectioned cartridges from each production lot (all void of charged delay body and powder charges) at 13 inches, using a steel ball 15/16 inches in diameter, weighing 1.94 ounces plus or minus 0.02 ounces, in a primer sensitivity machine with a cartridge holder approved by the procuring activity.

4.4.5 Sealant application.-The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length (360) of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

4.4.6 Hardness.-A hardness testing machine shall be used to determine the hardness values specified on the applicable drawing.

4.4.7 Functional.

4.4.7.1 Ballistics.-The sample for ballistic tests shall be grouped and temperature conditioned in accordance with 4.4.7.1.1 and rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample unit shall be reconditioned in accordance with 4.4.7.1.1.

4.4.7.1.1 Temperature conditioning.-A standard inert cartridge containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

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5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawing B8799620 and B8799621.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Delay time is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Loaded Assembly Hazard Information:
Explosive Hazard Classification: Class 3
Storage Compatibility: Group B
Department of Transportation (DOT) Hazard Class C
DOT Marking - Explosive Power Devices - Class C
Handle Carefully - (Keep Fire Away)

6.4 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive, consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

Project No. 1377- A506

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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